Contribution Journal – Module 7: Model Evaluation

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1. Introduction

In Lab 7, I learned how to evaluate a machine learning model beyond just checking its accuracy. I used the Titanic dataset and a Logistic Regression model to predict whether passengers would survive. This lab helped me understand how different evaluation methods can show how well a model performs and where it makes mistakes.

2. Specific Contribution

My contribution to this lab was preparing the Titanic dataset, training the Logistic Regression model, and writing Python code to evaluate its performance. I created a confusion matrix heatmap and generated a classification report to review precision, recall, and F1-score. My model achieved about 80% accuracy, and the five-fold cross-validation average was 78.68%. I also completed the knowledge check by explaining the differences between precision and recall and why cross-validation is useful. These activities helped me practice using Python and understand the results produced by a machine learning model.

3. Conclusion

Overall, this lab helped me improve my coding and model evaluation skills. I learned that accuracy alone does not explain everything about a model’s performance. Using a confusion matrix, precision, recall, F1-score, and cross-validation gave me a better understanding of how reliable the model was. I will use what I learned in future projects to evaluate models more carefully and make better decisions based on their results.